

Isolation Forest - Complete Notes

1. What is Anomaly Detection?

Anomaly detection is the process of identifying observations that significantly differ from the majority of the data. These unusual observations are called anomalies or outliers.

2. What is Isolation Forest?

Isolation Forest is an unsupervised machine learning algorithm used to detect anomalies. It works on the idea that anomalies are rare and different, therefore they can be isolated more quickly than normal observations.

3. Core Idea

Instead of profiling normal points, Isolation Forest isolates points using random splits. Anomalies require fewer splits to become isolated.

4. Step-by-Step Working

Step 1: Randomly select a feature.
Step 2: Randomly select a split value within the feature range.
Step 3: Split the data.
Step 4: Repeat until each observation is isolated.
Step 5: Record the path length (number of splits required).
Step 6: Build many such random trees.
Step 7: Compute the average path length across all trees.

5. Example

Data: Age = [25, 30, 35, 40, 120].

A random split at Age < 50 immediately isolates 120. Normal points require several additional splits. Therefore 120 has a shorter path length and is considered an anomaly.

6. Path Length Interpretation

Short Path Length -> Anomaly
Long Path Length -> Normal Observation

7. Why Does It Work?

Anomalies are few and far from dense regions. Random splits isolate them very quickly, while normal observations are embedded within clusters and need many splits.

8. Scikit-Learn Implementation

```
from sklearn.ensemble import IsolationForest
```

```
model = IsolationForest(contamination=0.05, random_state=42)  
model.fit(X)
```

```
predictions = model.predict(X)
```

Output:

1 = Normal

-1 = Anomaly

9. Important Parameters

n_estimators: Number of trees.

contamination: Expected percentage of anomalies.

max_samples: Samples used for each tree.

random_state: Reproducibility.

10. Advantages

- Fast and scalable.
- Works well with high-dimensional data.
- Does not require labeled data.
- Effective for large datasets.

11. Limitations

- Randomness may affect results slightly.
- Choosing contamination can be challenging.
- Less interpretable than simple statistical methods.

12. Interview Answer

Isolation Forest detects anomalies by constructing multiple random isolation trees. Since anomalies are rare and different from normal observations, they get isolated in fewer random splits and therefore have shorter average path lengths. Points with shorter average path lengths are classified as anomalies, while points with longer path lengths are considered normal.